Therefore, the expected toss of a fair six sided die is 3.5.

Example 6.18

Suppose the probability mass function of the discrete random variable is

$X = x$	0	1	2	3
$p(x)$	0.2	0.1	0.4	0.3

What is the value of $E(3X + 2X^2)$?

Solution:

$$E(X) = \sum_x x P_X(x)$$
$$= (0 \times 0.2) + (1 \times 0.1) + (2 \times 0.4) + (3 \times 0.3)$$
$$= 1.8$$
$$E\left(X^2\right) = \sum_x x^2 P_X(x)$$
$$= (0^2 \times 0.2) + (1^2 \times 0.1) + (2^2 \times 0.4) + (3^2 \times 0.3)$$
$$= 4.4$$
$$E(3X + 2X^2) = 3E(X) + 2E(X^2)$$
$$= (3 \times 1.8) + (2 \times 4.4)$$
$$= 14.2$$

Example 6.19

Consider a random variable X with probability density function

$$f(x) = \begin{cases} 4x^3, if\ 0 < x < 1 \\ 0, \ otherwise \end{cases}$$

Find $E(X)$ and $V(X)$.

Solution:

We know that,

$$E(X) = \int_{-\infty}^{\infty} x f(x)dx$$
$$= \int_0^1 x\, 4x^3\, dx$$
$$= 4\left[\frac{x^5}{5}\right]_0^1$$
$$E(X) = \frac{4}{5}$$
$$E\left(X^2\right) = \int_{-\infty}^{\infty} x^2 f(x)dx$$
$$= \int_0^1 x^2\, 4x^3\, dx$$
$$= 4\left[\frac{x^6}{6}\right]_0^1$$
$$= \frac{4}{6} = \frac{2}{3}$$
$$V(X) = E\left(X^2\right) - \left[E\left(X\right)\right]^2$$
$$= \frac{4}{6} - \left[\frac{4}{5}\right]^2$$
$$= \frac{2}{75}$$

Example 6.20

If $f(x)$ is defined by $f(x) = ke^{-2x}, 0 \le x < \infty$ is a density function. Determine the constant k and also find mean.

Solution:

We know that

$\int_{-\infty}^{\infty} f(x)dx = 1$, since $f(x)$ is a density function.

$$\int_0^{\infty} k e^{-2x}\, dx = 1$$
$$k\int_0^{\infty} e^{-2x}\, dx = 1$$
$$k\left[\frac{e^{-2x}}{-2}\right]_0^{\infty} = 1$$
$$\Rightarrow k = 2$$
$$E(X) = \int_{-\infty}^{\infty} x f(x)dx$$
$$= \int_0^{\infty} xk e^{-2x}\, dx = 2\int_0^{\infty} xe^{-2x}\, dx$$
$$= 2\left\{\left[\frac{xe^{-2x}}{-2}\right]_0^{\infty} - \int_0^{\infty} \frac{e^{-2x}}{-2} dx\right\}$$
$$\left(\because \int udv = uv - \int vdu\right)$$
$$= \int_0^{\infty} e^{-2x}\, dx = \frac{1}{2}$$

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